

Sample questions – Module 1 (Source Elmasri & Navathe)

1. Define the following terms: *data, database, DBMS, database system, database catalog, program-data independence, user view, DBA, end user, canned transaction.*
2. Discuss the main characteristics of the database approach and how it differs from traditional file systems.
3. What are the responsibilities of the DBA and the database designers?
4. What are the different types of database end users? Discuss the main activities of each.
5. Discuss the components of typical database management system software.
6. When would you not use a DBMS?
7. Define the following terms: *data model, database schema, database state, internal schema, conceptual schema, external schema, data independence, DDL, DML, query language, host language, data sublanguage, client/server architecture, three-tier architecture.*
8. Discuss the main categories of data models.
9. What is the difference between a database schema and a database state?
10. Describe the three-schema architecture.
11. What is the difference between logical data independence and physical data independence? Which one is harder to achieve? Why?
12. What is the difference between the two-tier and three-tier client/server architectures?